
Learning to Defer for Physics-Informed Neural Networks

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Abstract

1 Physics-informed neural networks (PINNs) provide cheap, mesh-free, differentiable
2 surrogates for partial differential equations, but their pointwise accuracy is known
3 to be *spatially non-uniform*: a small subdomain typically carries a disproportionate
4 share of the error. Existing remedies—adaptive sampling, loss reweighting, causal
5 training, domain decomposition—rebalance the optimisation but still return a single
6 global predictor that the user must trust everywhere. We argue instead for *learning*
7 *with abstention*: a frozen PINN is paired with a classical solver, and a learned
8 rejector decides at each spatial location whether to commit to the PINN prediction
9 or to abstain to the solver. We cast this as a regression abstention problem with an
10 input-dependent cost derived from the PDE residual and a linearised error-transport
11 estimate, derive its Bayes-optimal rejector, and design a consistent surrogate for
12 the predictor–rejector formulation extended from a categorical predictor with 0–1
13 loss to a continuous predictor with a non-negative discrepancy. We prove an $\mathcal{H}$ -
14 consistency bound for the proposed surrogate and show that the induced hybrid
15 solution inherits a controllable PDE error from the rejection decisions.

16 1 Introduction

17 Physics-informed neural networks (PINNs) are a widely used deep-learning surrogate for partial
18 differential equations (Cuomo et al., 2022; Karniadakis et al., 2021; Lu et al., 2021; Raissi et al.,
19 2019). Their appeal is clearest after training: a neural network represents a mesh-free, differentiable
20 field that can be evaluated at arbitrary spatial or space-time locations without running a new global
21 linear or nonlinear solve (Brenner and Scott, 2008; LeVeque, 2007). This amortised evaluation
22 cost is attractive, but it does not by itself provide the local accuracy guarantees expected of a
23 resolved numerical discretisation. On several standard PDE benchmarks, PINNs can exhibit large
24 errors in regimes with multiscale structure, stiffness, sharp gradients, or unbalanced residual terms
25 (Krishnapriyan et al., 2021; Wang et al., 2021, 2022b). Existing a-posteriori analyses relate residual
26 norms to global solution errors under stability and approximation assumptions (Mishra and Molinaro,
27 2023); such bounds are valuable, but they do not directly identify the particular points or subregions
28 where a trained PINN should be trusted.

29 The missing object is therefore a *local trust rule*. Empirically, PINN errors are often highly non-
30 uniform over the domain: a model may approximate large smooth regions well while failing near
31 fronts, boundary layers, singular features, or other regions whose solution components are difficult for
32 the training dynamics to capture (Krishnapriyan et al., 2021; Wang et al., 2022a,b). This observation
33 suggests a different deployment goal from uniform replacement of a classical solver. Instead of
34 asking a single trained PINN to be accurate everywhere, we ask it to predict where it is accurate
35 enough to use and where it should abstain. The natural statistical language for this goal is learning
36 with abstention (Bartlett and Wegkamp, 2008; Cao et al., 2022; Chow, 1970; Cortes et al., 2016,

2024; Geifman and El-Yaniv, 2017; Mao et al., 2024a,g; Ni et al., 2019): a predictor is paired with a rejector that decides, for each input, whether to return the prediction or abstain.

Classical abstention would return only a rejection symbol $\perp$ on rejected inputs. For PDE surrogates this is usually not enough, because downstream simulation, visualisation, or physical diagnostics often require a complete solution field. We therefore study abstention with a fallback solver: a frozen PINN $\hat{u}_\theta$ supplies the field on accepted points, while a classical numerical solver is used on rejected regions or queried locations. If the rejector abstains on a small fraction of the domain, the hybrid can preserve much of the amortised efficiency of the PINN while using the solver where the PINN is estimated to be unreliable. The precise cost depends on the implementation of the fallback solve and the geometry of the rejected set, so we model it abstractly through an input-dependent abstention cost $c(x)$ rather than assuming a universal per-point solver cost.

This perspective is complementary to the large body of work that improves PINN training. Loss reweighting based on gradient-flow or neural-tangent kernel diagnostics (Wang et al., 2021, 2022b), self-adaptive collocation weights (McClenny and Braga-Neto, 2023), causal residual training (Wang et al., 2022a), residual-based adaptive sampling (Daw et al., 2023; Wu et al., 2023), and domain decomposition methods (Jagtap and Karniadakis, 2020; Jagtap et al., 2020; Moseley et al., 2023) can substantially improve particular classes of PDE problems. However, they still return a single global predictor. They do not, by themselves, convert a trained PINN into a pointwise certificate of reliability, nor do they give a post-hoc rule for routing an already-trained model to a fallback solver. Our goal is to add precisely that missing layer: a learned rejector whose decision is calibrated to the local error of a fixed PINN relative to the cost of abstention.

In the setting studied here, the trained PINN is fixed and the learned object is only the router. This makes the problem simpler than joint predictor–rejector learning, but different from standard classification with rejection. At each point, the router compares two continuous costs: the estimated cost of using the PINN, represented by a non-negative residual–error score $R(x)$, and the cost of invoking the classical solver, denoted by β . The Bayes router is therefore a level-set rule: use the solver exactly where $R(x) > \beta$. The theoretical task is to show that the smooth logistic router loss used for training is calibrated for this hard threshold rule, and that small surrogate excess implies small excess routing cost after thresholding.

Contributions.

- We formulate **learning with abstention for PINNs**: a frozen PINN $\hat{u}_\theta$ is paired with a fallback numerical solver, and a learned router decides pointwise whether to use the PINN or defer to the solver.
- We define the PINN-side routing cost $R(x)$ from the local PDE residual and a linearised error-transport estimate, and compare it with a solver cost β .
- We derive the Bayes-optimal router as the threshold rule $\mathbb{1}\{R(x) > \beta\}$ and prove that the logistic router surrogate is calibrated for this decision rule, with an $\mathcal{R}$ -consistency bound.
- We analyse the hybrid solution induced by the router partition, including the one-way Dirichlet coupling at the PINN–solver interface.

2 Background

We define for the rest of the paper: learning to reject for a fixed regression rule (Section 2.1) and the standard PINN formulation for PDEs (Section 2.2).

2.1 Learning to reject in regression

Let (X, Y) be a random pair with joint distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$, where $\mathcal{X} \subseteq \mathbb{R}^{d_x}$ is a measurable input space and $\mathcal{Y} \subseteq \mathbb{R}^{d_y}$ is a measurable response space. Throughout, $\psi : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}$ denotes a measurable non-negative discrepancy, such as the squared loss $\psi(a, b) = \frac{1}{2}\|a - b\|_2^2$; boundedness of ψ is assumed only when it is required by a stated consistency bound.

A regression rule with rejection is a pair (h, r) in which $h : \mathcal{X} \rightarrow \mathcal{Y}$ is a measurable predictor and $r : \mathcal{X} \rightarrow \{0, 1\}$ is a measurable rejector. We use the convention $r(x) = 0$ to accept the prediction

86 $h(x)$ and $r(x) = 1$ to abstain. Following the classical rejection literature, abstention trades prediction
 87 error against a rejection cost (Bartlett and Wegkamp, 2008; Chow, 1970; Cortes et al., 2016, 2024).
 88 Most standard formulations take this cost to be constant, i.e. $c(x) \equiv c$ (Bartlett and Wegkamp, 2008;
 89 Cao et al., 2022; Chow, 1970; Geifman and El-Yaniv, 2017; Mao et al., 2024a; Ni et al., 2019); we
 90 write $c : \mathcal{X} \rightarrow [0, C]$ to allow the more general input-dependent case, expressed in the same units as
 91 ψ . We work in a second-stage predictor–rejector setting (Cheng et al., 2023; Cortes et al., 2016; Mao
 92 et al., 2024a), where h is fixed and only r is learned. The pointwise loss is

$$\ell_{\text{abs}}(h, r; x, y) := \psi(h(x), y) \mathbb{1}\{r(x) = 0\} + c(x) \mathbb{1}\{r(x) = 1\}, \quad (1)$$

93 with population risk $\mathcal{E}_{\ell_{\text{abs}}}(h, r) := \mathbb{E}_{(X, Y) \sim \mathcal{D}}[\ell_{\text{abs}}(h, r; X, Y)]$. When the rule accepts, it pays the
 94 discrepancy of the fixed predictor; when it abstains, it pays $c(X)$.

95 **Lemma 1** (Bayes-optimal rejector). *For a fixed predictor h , define $\eta_h(x) := \mathbb{E}[\psi(h(x), Y) \mid X = x]$.
 96 Then, for $\mathbb{P}_X$ -a.e. x , the Bayes-optimal rejector is*

$$r^*(x) \in \begin{cases} \{0\}, & \eta_h(x) \leq c(x), \\ \{1\}, & \eta_h(x) > c(x), \end{cases}$$

97 where $r^*(x) = 0$ accepts $h(x)$ and $r^*(x) = 1$ abstains.

98 The proof is the pointwise comparison of the two available costs, $\eta_h(x)$ and $c(x)$. Thus the rejector
 99 only needs to learn the sign of $\eta_h(x) - c(x)$. We use a measurable score $s : \mathcal{X} \rightarrow \mathbb{R}$ from a rejector
 100 class $\mathcal{H}_r$ and define $r_s(x) := \mathbb{1}\{s(x) > 0\}$. A smooth surrogate Φ_{abs} for the discontinuous loss (1)
 101 is useful only if small surrogate excess risk controls the true abstention excess risk in $\mathcal{H}_r$, in the sense
 102 of $\mathcal{H}$ -consistency bounds (Awasthi et al., 2022; Mao et al., 2023b, 2024a). This abstract setup is the
 103 only learning-to-reject machinery needed in the sequel.

104 2.2 Physics-informed neural networks

105 Let $\Omega \subset \mathbb{R}^d$ be a bounded open domain with Lipschitz boundary $\partial\Omega$, and consider the boundary-value
 106 problem

$$\mathcal{N}[u](x) = f(x), \quad x \in \Omega, \quad \mathcal{B}[u](x) = g(x), \quad x \in \partial\Omega, \quad (2)$$

107 where $\mathcal{N}$ is a possibly nonlinear differential operator, $\mathcal{B}$ is a boundary operator, and f, g are given
 108 data in function spaces appropriate for the problem. For the strong-form notation used here, we
 109 assume (2) admits a unique sufficiently smooth solution $u^* : \bar{\Omega} \rightarrow \mathbb{R}^{d_u}$, with enough regularity for
 110 the operators below to be evaluated pointwise.

111 A PINN is a parameterised map $\hat{u}_\theta : \Omega \rightarrow \mathbb{R}^{d_u}$, typically a feed-forward neural network with
 112 differentiable activations, trained so that the PDE and boundary residuals are small. The residuals are

$$\mathbf{r}_{\text{int}}^\theta(x) := \mathcal{N}[\hat{u}_\theta](x) - f(x), \quad \mathbf{r}_{\text{bd}}^\theta(x) := \mathcal{B}[\hat{u}_\theta](x) - g(x), \quad (3)$$

113 and are computed by automatic differentiation through $\hat{u}_\theta$ when the operators are represented in
 114 strong form (Karniadakis et al., 2021; Lu et al., 2021; Raissi et al., 2019). Given interior collocation
 115 points $\{x_i^{\text{int}}\}_{i=1}^{N_{\text{int}}} \subset \Omega$ and boundary collocation points $\{x_j^{\text{bd}}\}_{j=1}^{N_{\text{bd}}} \subset \partial\Omega$, a standard PINN objective
 116 is

$$\mathcal{L}_{\text{PINN}}(\theta) := \frac{\lambda_{\text{int}}}{N_{\text{int}}} \sum_{i=1}^{N_{\text{int}}} \|\mathbf{r}_{\text{int}}^\theta(x_i^{\text{int}})\|^2 + \frac{\lambda_{\text{bd}}}{N_{\text{bd}}} \sum_{j=1}^{N_{\text{bd}}} \|\mathbf{r}_{\text{bd}}^\theta(x_j^{\text{bd}})\|^2, \quad (4)$$

117 with non-negative weights λ_{int} and λ_{bd} . Data-fit terms on observed solution values can be added
 118 analogously.

119 A central difficulty is that (4) directly constrains residuals only through sampled empirical averages,
 120 whereas the deployment quantity of interest is often the local solution error $\hat{u}_\theta(x) - u^*(x)$. A-
 121 posteriori results connect these quantities at the level of global norms. Schematically, under suitable
 122 stability, regularity, approximation, and sampling assumptions, Mishra and Molinaro (2023) obtain
 123 bounds of the form

$$\|\hat{u}_\theta - u^*\|_{L^2(\Omega)} \leq C_{\mathcal{N}} \|\mathbf{r}_{\text{int}}^\theta\|_{L^2(\Omega)} + C_{\mathcal{B}} \|\mathbf{r}_{\text{bd}}^\theta\|_{L^2(\partial\Omega)} + \text{approximation/sampling terms}, \quad (5)$$

124 where the constants and extra terms depend on the PDE class, the domain, and the sampling scheme.
 125 Equation (5) should therefore be read as a global stability statement, not as a pointwise certificate of

trust. The abstention layer introduced in Section 3 addresses this local question by taking the fixed predictor h in Section 2.1 to be the frozen PINN $\hat{u}_\theta$, whether it was trained by (4) or by modern variants such as adaptive weighting, causal training, adaptive sampling, or domain decomposition (Daw et al., 2023; Jagtap and Karniadakis, 2020; Jagtap et al., 2020; McClenny and Braga-Neto, 2023; Moseley et al., 2023; Wang et al., 2021, 2022a; Wu et al., 2023).

3 Bayes Rule and Consistent Surrogate

(Placeholder. To be developed in the next pass.)

This section will (i) instantiate the abstention loss (1) of Section 2.1 with $h = \hat{u}_\theta$ and an input-dependent cost derived from the residual of Section 2.2, (ii) derive the Bayes-optimal rejector as a threshold rule on the conditional risk $\eta_h(x)$ relative to the solver cost, in the spirit of Mao et al. (2024a) but for a continuous predictor, and (iii) prove an $\mathcal{H}$ -consistency bound for the proposed surrogate using the comp-sum machinery of Mao et al. (2023b, 2024f).

4 Experiments

(Placeholder.)

We will validate the framework on (i) a Poisson problem with manufactured sharp features that target the known PINN failure modes (Krishnapriyan et al., 2021; Wang et al., 2022b); (ii) steady Navier–Stokes flow past a cylinder; and (iii) a generalisation suite testing transfer across geometries and across PINNs trained on mismatched configurations.

5 Conclusion

(Placeholder.) We introduced learning-to-defer for physics-informed neural networks: a frozen PINN is paired with classical solvers and a learned router that decides, pointwise, which agent to trust. The framework inherits the Bayes-consistency and $\mathcal{H}$ -consistency apparatus of the L2D literature while addressing a structurally different prediction problem.

Limitations

(Placeholder.) The framework assumes (i) availability of a reference solution oracle for training, possibly via held-out high-fidelity solver runs; (ii) frozen PINN and frozen experts, deferring online and joint adaptation to future work; and (iii) costs $\beta_j(x)$ that are known or estimable at deployment.

Impact Statement

(Placeholder.) Reliable surrogate models for PDEs underpin scientific computing across climate, fluids, and materials. A principled deferral layer mitigates the silent-failure risk of trusting a PINN globally, but it inherits the biases of its constituent solvers and of the cost structure chosen by the operator. Deployment in safety-critical scientific pipelines should validate the deferral policy against domain-expert oracles.

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292 A Appendix

293 (Placeholder for proofs, additional experiments, and notation table.)